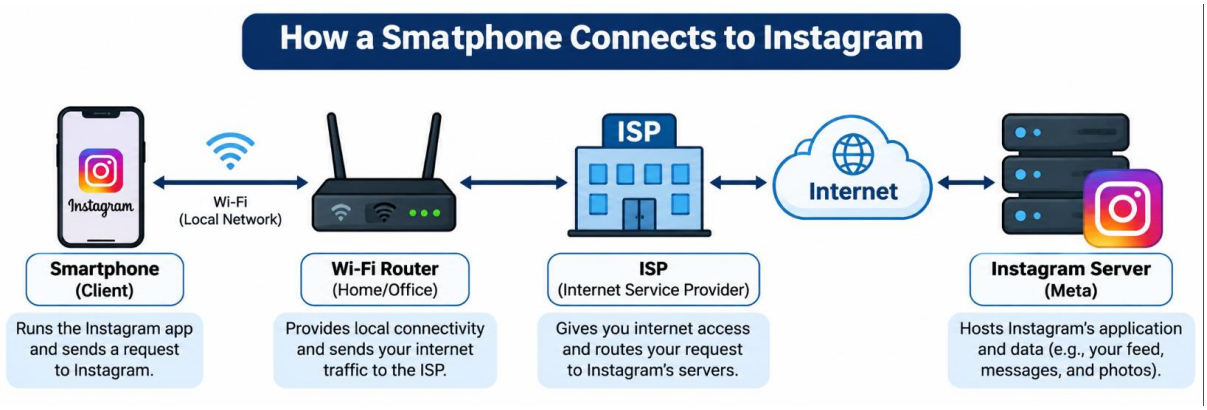
Networking Technologies

Task – 1 : Draw a simple network diagram showing how your smartphone connects to Instagram — include your device, Wi-Fi router, ISP, and Instagram's server, and label each part.

Solution : Smartphone to Instagram Server



Explanation of the diagram:

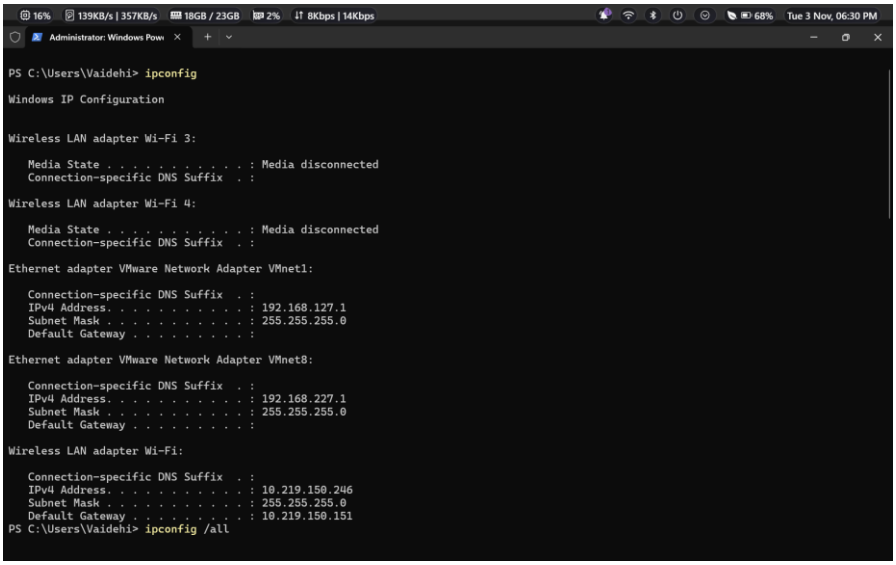
1. Smartphone: The user opens Instagram on their mobile phone.
2. Wi-Fi Router: Connects the smartphone to the local wireless network and forwards data.
3. ISP (Internet Service Provider): Provides internet connectivity and routes data towards Instagram.
4. Instagram Server: Receives requests, processes them, and sends information such as posts, reels, and messages back to the smartphone.

Data Flow:

Smartphone → Wi-Fi Router → ISP → Internet → Instagram Server

Task – 2 : List the IP address of your device and your default gateway using the 'ipconfig' command (Windows) or 'ifconfig' (Linux/Mac), and explain what each means in 1-2 lines.

Solution : Command for Windows



Network Detail	**Example**	**Meaning**
IP Address	10.219.150.246	A unique local address assigned to my device within the network.
Default Gateway	10.219.150.151	The router address through which my device communicates with other networks and the internet.

Task – 3 : Simulate a Zomato food order journey by writing the steps/data packets that travel from your device to Zomato's server and back, mentioning protocols like HTTP, TCP/IP, and DNS wherever relevant. Hint : Think about what happens when you search for a restaurant and place an order.

Solution : Ordering food through Zomato

When I order food using the Zomato application, several networking steps take place between my smartphone and Zomato's server.

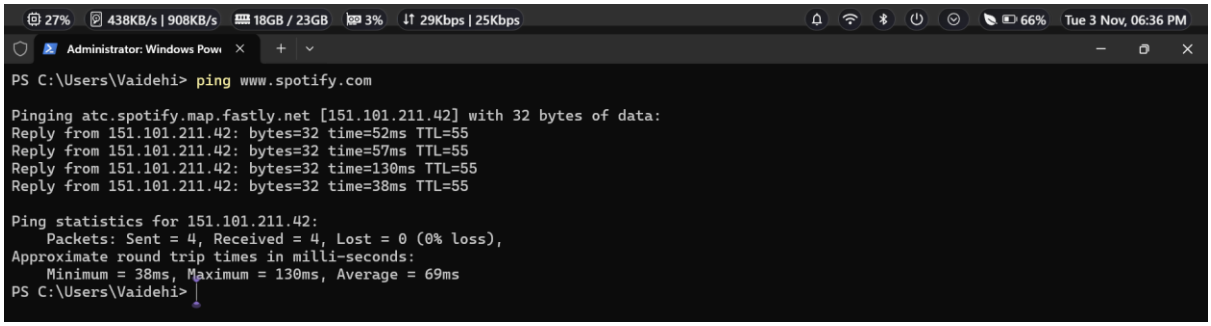
Step-by-step communication:

1. Open Zomato: I open the Zomato application on my smartphone and connect to the internet using Wi-Fi or mobile data.
2. DNS Resolution: My device uses DNS (Domain Name System) to translate Zomato's domain name into an IP address so it can locate the server.
3. TCP Connection: My device establishes a connection with the server using TCP (Transmission Control Protocol), which provides reliable data transmission.
4. HTTPS Request: The application sends an encrypted request to Zomato's server using HTTPS. HTTPS uses TLS encryption to protect the communication.
5. Restaurant Search: When I search for a restaurant, my device sends the search request to Zomato's server.
6. Server Response: Zomato's server processes the request and sends restaurant details, menus, prices, and availability back to my device.
7. Place Order: I select food items and confirm my order. The application sends the order details to the server through another HTTPS request.
8. Order Confirmation: The server processes the order and sends an order confirmation and tracking information back to my device.
9. Protocols Used:

Protocol	Purpose
DNS	Converts domain names into IP addresses.
TCP	Provides reliable communication between devices and servers.
IP	Handles addressing and routing of data packets.
HTTPS	Provides secure, encrypted communication between the application and server.
TLS	Encrypts communication and helps authenticate the server.

Task – 4 : Use the 'ping' command to check the connection between your computer and www.spotify.com, then explain what the results mean for network troubleshooting.

Solution :



How Ping Helps in Troubleshooting:

1. Successful Reply: Indicates that the destination is responding to ICMP requests.
2. High Response Time: May indicate network congestion or latency.
3. Request Timed Out: May indicate connectivity problems or that the destination is blocking ICMP requests.
4. 100% Packet Loss: Indicates that no replies were received, but it does not necessarily mean the website is unavailable.